

Why is the odds ratio the only appropriate parameter if the sampling is retrospective?

The smoking and lung cancer data, for example, cannot be used to estimate the individual proportions of smokers and nonsmokers who get lung cancer or the difference between the proportions. That is because the way they have been obtained. However, the odds ratio is the same regardless of which factor is considered the response. Let's illustrate this issue with an example.

Smoking status - Cancer Status	Lung Cancer	No Cancer	Total
Smokers	688	650	1338
Nonsmokers	21	59	80
Total	709	709	1418

Defining $Y = \text{cancer status}$ as the response variable and $X = \text{smoking status}$ as the explanatory variable. Odds of cancer among smokers ($\hat{p}_1$ is the estimated probability of success (cancer) in the group of smokers):

$$\text{odds}_{X=1} = \frac{\hat{p}_1}{1 - \hat{p}_1} = \frac{P(Y = 1|X = 1)}{1 - P(Y = 1|X = 1)} = \frac{P(Y = 1|X = 1)}{P(Y = 0|X = 1)} = \frac{688/1338}{650/1338} = \frac{688}{650}$$

Odds of cancer among nonsmokers ($\hat{p}_2$ is the estimated probability of success (cancer) in the group of smokers):

$$\text{odds}_{X=0} = \frac{\hat{p}_2}{1 - \hat{p}_2} = \frac{P(Y = 1|X = 0)}{1 - P(Y = 1|X = 0)} = \frac{P(Y = 1|X = 0)}{P(Y = 0|X = 0)} = \frac{21/80}{59/80} = \frac{21}{59}$$

Odds ratio for cancer relative to being a smoker (for cancer being a smoker relative to being a nonsmoker). It is the ratio of the previous two:

$$\hat{\theta} = \frac{688 \cdot 59}{650 \cdot 21}$$

Now, changing the role of the response and the explanatory variable. Now, being a smoker is defined as the *success*. Odds of smoker among cancer patients:

$$\frac{P(X = 1|Y = 1)}{1 - P(X = 1|Y = 1)} = \frac{P(X = 1|Y = 1)}{P(X = 0|Y = 1)} = \frac{688/709}{21/709} = \frac{688}{21}$$

Odds of smoker among noncancer patients:

$$\frac{P(X = 1|Y = 0)}{1 - P(X = 1|Y = 0)} = \frac{P(X = 1|Y = 0)}{P(X = 0|Y = 0)} = \frac{650/709}{59/709} = \frac{650}{59}$$

Odds ratio for smoker relative to being a cancer patient (for smoker being a cancer patient relative to being a patient without cancer). It is the ratio of the previous two:

$$\hat{\theta} = \frac{\text{odds}_{X=1}}{\text{odds}_{X=0}} = \frac{688 \cdot 59}{650 \cdot 21}$$

Consequently, the odds ratio can be estimated either by observing cancer incidence in independent samples of smokers and nonsmokers or by observing proportions of smokers in independent samples of cancer victims and controls. The odds ratio is the only quantity pertaining to the prospective population that can be estimated from retrospective studies.